Operations Research

Nadia Brauner





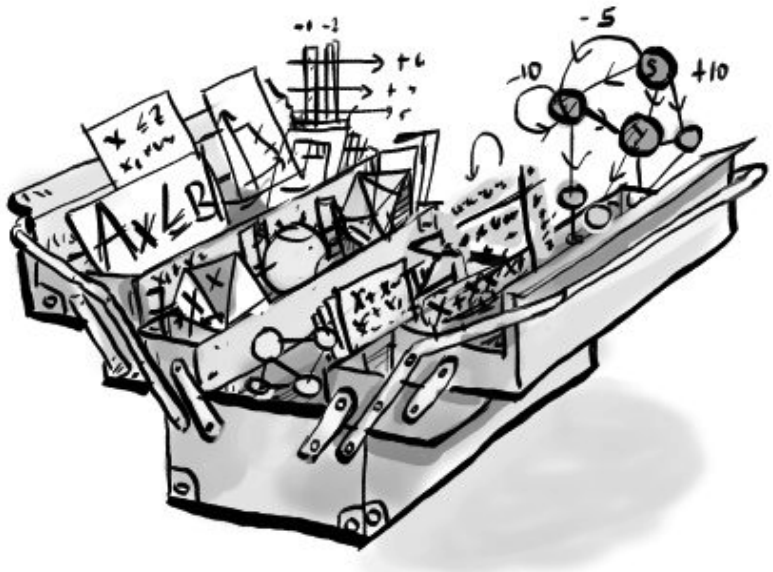
Operations Research

- Scientific methods for solving decision problems in organizations
- A toolbox of methods, techniques, algorithms, theorems, etc. to solve optimization problems

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Operations Research:
the science of better



dessin de Lionel Lagarde

OR in your everyday life

- 08:00: optimising the collection of waste recycling
- ...
- 15:00: relocation of shared bicycles
- 17:00: managing public transport delays to minimize the impact on passengers
- ...

Operations Research

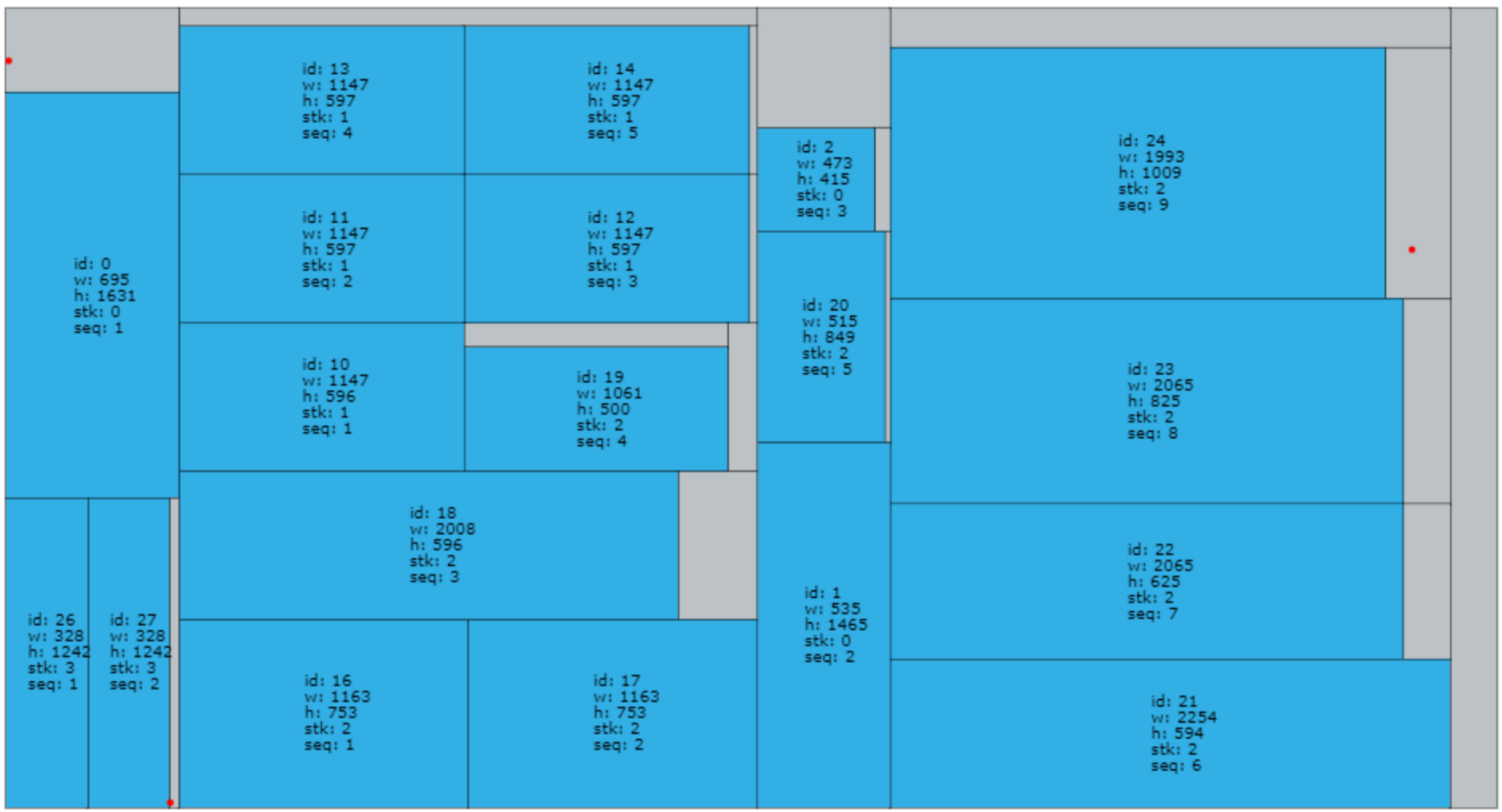
The Roadef/Euro challenges `http://challenge.roadef.org/`

2022	3D truck loading optimization	(Renault)
2020	Maintenance planning Problem	(RTE)
2018	A cutting optimization problem	(Saint Gobain)
2016	An inventory routing problem	(Air Liquide)
2014	Trains don't vanish !	(SNCF)
2012	Machine reassignment	(Google)
2010	A large-scale energy management problem	(EDF)
2009	Disruption management for commercial aviation	(Amadeus)
2007	Technicians and interventions scheduling	(France Telecom)
2005	A car sequencing problem	(Renault)
2003	Missions of earth observation satellites	(ONERA et CNES)
2001	Frequency assignment problem	(CELAR, army)
1999	Inventory Management problem	(Bouygues)

SAINT-GOBAIN

The Roadef/Euro challenges 2018

- Glass cutting optimization



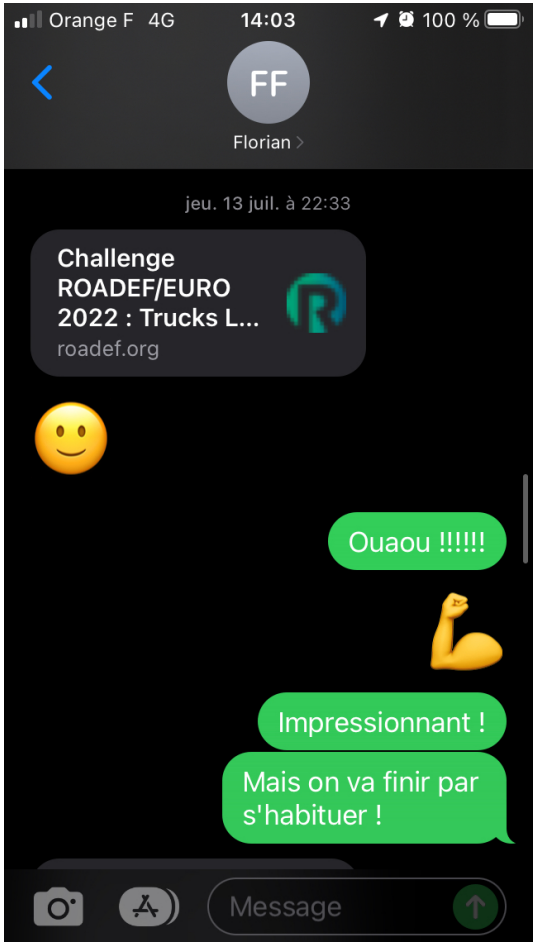
The challenge ROADEF/EURO 2018

- Glass cutting optimization



Recherche Opérationnelle: Applications

The Roadef/Euro challenges 2022



during the fireworks

Recherche Opérationnelle

OSUG: Scheduling **observations of celestial objects**

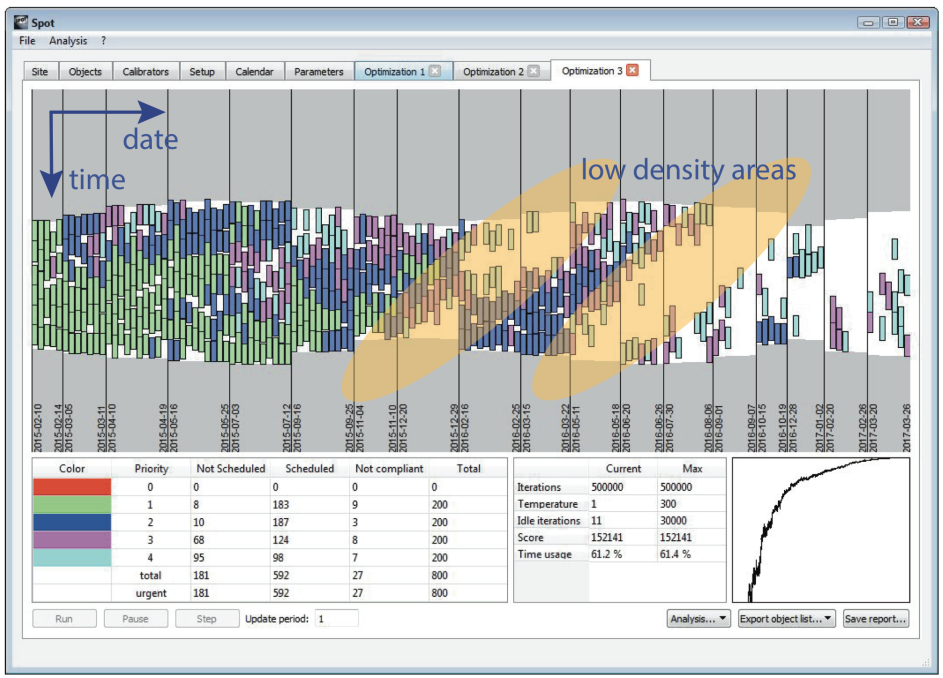


photo credits: Spot

Looking for exoplanets:

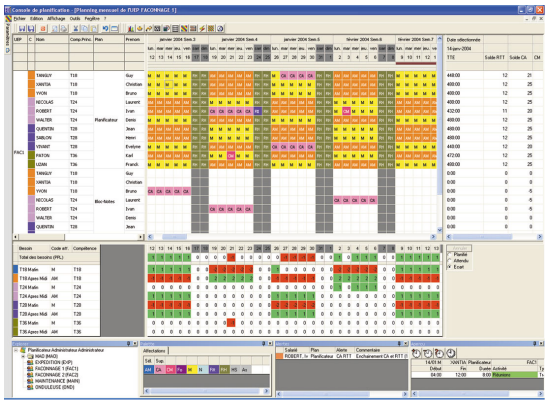
`https://youtu.be/kFpYprqJgUQ`

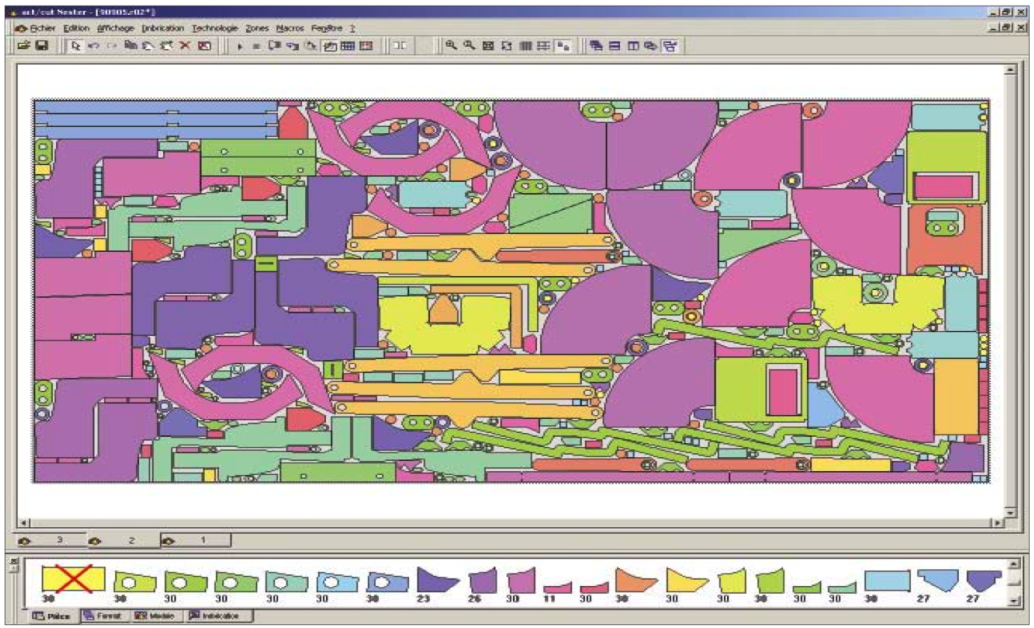
Operations Research



Optimization methods

- Model a practical problem
- Solve (algorithms)
- Use/analyze the results





Modelization of a problem of marine spatial planning

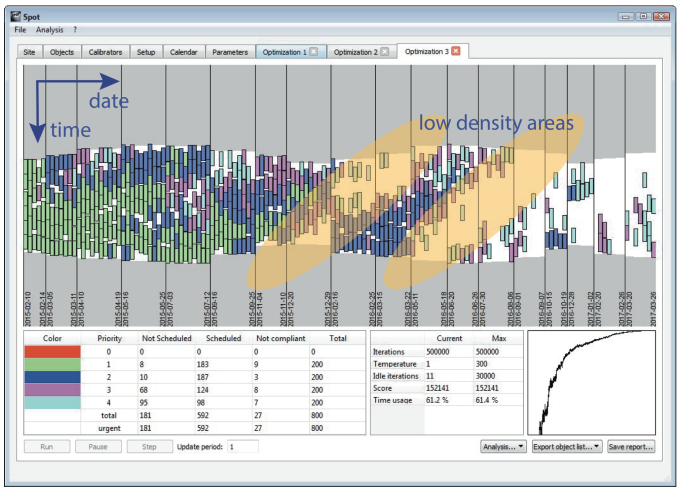
$b_i = 1$ the species i is protected, 0 otherwise, $i = 1, 2...p$
$x_j = 1$ if area j is protected, 0 otherwise, $j = 1, 2...n$

$\max \sum_{i=1}^{p} b_i$		max number of protected species
$\sum_{j=1}^{n} \mathbf{p_j} x_j \leq \mathbf{B}$		budget constraint
$\mathbf{r_i} b_i \leq \sum_{j=1}^{n} \mathbf{a_{ij}} x_j$	$\forall i = 1, 2...p$	the species is at least in $\mathbf{r_i}$ protected areas

Operations Research

This course

- Linear Programming
  - Simplex method, duality
- Mixed Integer Programming
  - Combinatorial optimization, modeling technics, Branch and bound, Industrial language
- Dynamic Programming
  - From theory to practice

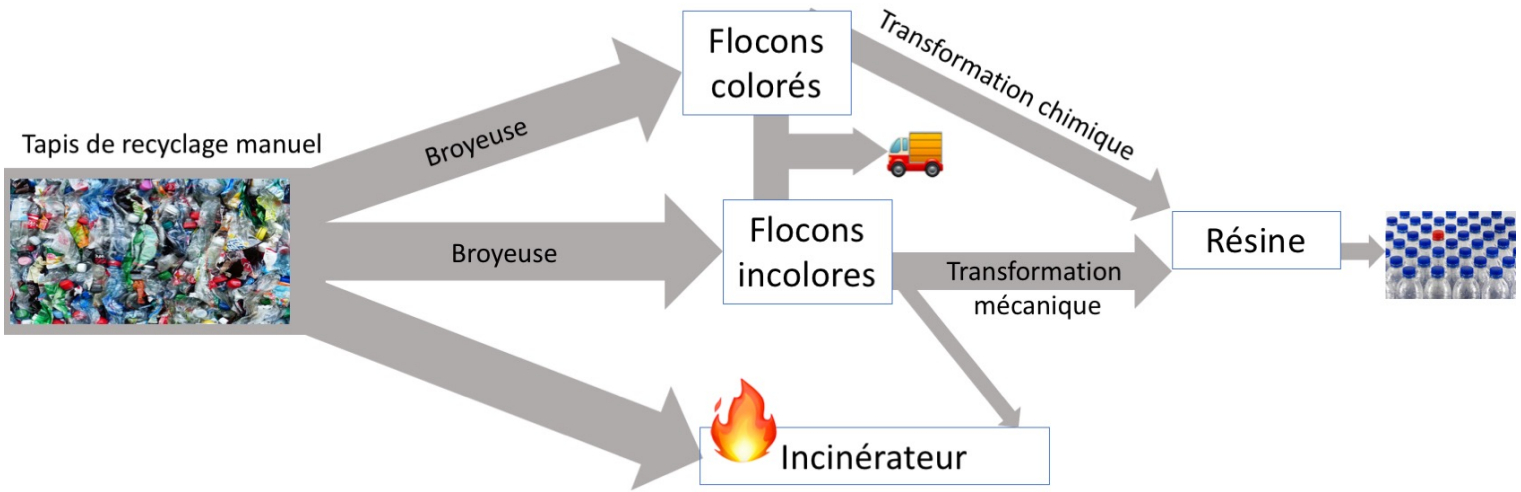




Recherche Opérationnelle

At the end of this course, you will be able to

- Model a problem of biodiversity preservation
- Help in the dimensioning of a hydro-electric power plant
- Help local farmers organize their harvesting activities
- Organize the recycling of plastic waste...



For further information

- caseine.org
  - **Operations Research (UGA)**
  - Jeu de piste en Graphistan (in French)
  - Graphes (UGA)
- Master 2 ORCO



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